

Step-by-Step Guide: Analysing Traffic with Wireshark (GUI and CLI)

Wireshark is the de facto standard for network packet analysis. This guide covers both interfaces: the graphical **Wireshark GUI** and the command-line **tshark** tool that ships with it. In the CREST CPIA lab, tshark runs inside the `cpia-forensics` analyst container; the GUI runs on a desktop with a display.

The worked examples below use the real CPIA lab capture at `/work/network.pcap` (also at `~/labs/cpia-labs/work/network.pcap` on the host). This capture contains the actual incident: an SSH brute force and HTTP reconnaissance from attacker host `172.19.0.4` against victim `172.19.0.3`.

1. Understand the two tools

Tool	Interface	Use
Wireshark	GUI	Interactive capture, protocol dissection, follow streams
tshark	CLI	Scriptable capture and analysis, offline pcap reading
capinfos	CLI	Metadata about a capture file
mergcap	CLI	Merge multiple capture files
editcap	CLI	Split / filter capture files by time or packet

Both `wireshark` and `tshark` use the **same display filter syntax**, so anything you learn in one transfers to the other.

2. Start a capture

GUI (wireshark)

1. Open the Capture panel (or `Ctrl+K`).
2. Select the interface you want to monitor (e.g. `eth0`).
3. Click the blue shark-fin button, or press `Ctrl+E` to begin.
4. Click the red stop button (or `Ctrl+E`) to end and keep the capture.

CLI (tshark)

List the available interfaces:

```
tshark -D
```

Capture live on an interface, writing to a file:

```
tshark -i eth0 -w /work/live.pcap
```

Capture with a limit (e.g. 100 packets) and stop:

```
tshark -i eth0 -c 100 -w /work/live100.pcap
```

In the CPIA lab you normally analyse an existing capture rather than capture live, because the lab traffic was already collected during the incident.

3. Read an existing capture

GUI

File > Open, then select the .pcap file. The three panes appear:

- **Packet list** (top) - each captured frame on one row
- **Packet details** (middle) - the decoded protocol tree for the selected packet
- **Packet bytes** (bottom) - the raw hexadecimal and ASCII view

CLI

Read the capture and print a one-line summary per packet:

```
tshark -r /work/network.pcap
```

Count how many packets are in the file:

```
tshark -r /work/network.pcap 2>/dev/null | wc -l  
# 450 packets
```

Get capture file metadata:

```
capinfos /work/network.pcap
```

4. Understand what is in the capture

Before filtering, see which protocols dominate. The protocol hierarchy shows every protocol layer and how much data it carries:

```
tshark -r /work/network.pcap -q -z io,phs 2>/dev/null
```

For the CPIA lab capture this shows:

frame	frames:440	
eth	frames:440	
ip	frames:434	
tcp	frames:434	
ssh	frames:99	# brute force
http	frames:36	# reconnaissance

Which tells you the story at a glance: an SSH attack and HTTP probing.

See the two hosts involved (the attacker and victim):

```
tshark -r /work/network.pcap -q -z endpoints,ip 2>/dev/null
```

5. Filter with display filters

Display filters narrow what you see to packets matching a condition. They work identically in the GUI filter bar and in tshark -Y.

Common filters

```
ip.addr == 172.19.0.4      # all traffic to/from the attacker
ip.src == 172.19.0.4      # traffic FROM the attacker
ip.dst == 172.19.0.3      # traffic TO the victim
tcp.port == 22            # SSH
tcp.port == 80            # HTTP
http.request              # only HTTP request packets
http.response             # only HTTP responses
ssh                       # all SSH traffic
```

CLI examples (using -Y)

```
# all packets involving the attacker
tshark -r /work/network.pcap -Y "ip.addr == 172.19.0.4" 2>/dev/null

# only HTTP requests
tshark -r /work/network.pcap -Y "http.request" 2>/dev/null
```

That HTTP filter reveals the attacker's reconnaissance path:

```
226 4.03 GET /admin HTTP/1.1
238 4.14 GET /admin/index.php HTTP/1.1
251 4.25 GET /config.php HTTP/1.1
263 4.35 GET /backup HTTP/1.1
275 4.46 GET /backup.zip HTTP/1.1
287 4.57 GET /phpmyadmin/ HTTP/1.1
299 4.68 GET /wp-login.php HTTP/1.1
311 4.79 GET /server-status HTTP/1.1
```

This is textbook reconnaissance: probing admin panels, backup files, database front-ends, and status pages.

GUI

Type the same expression into the green filter bar (top of the window) and press Enter. Use the book icon to save reusable filters.

6. Combine and refine filters

Use Boolean operators to build precise queries:

```
ip.addr == 172.19.0.4 && tcp.port == 80      # attacker's HTTP traffic
http.request && ip.src == 172.19.0.4         # requests sent by attacker
tcp.port == 22 || tcp.port == 80            # SSH or HTTP
!(arp)                                       # everything except ARP
```

```
tshark -r /work/network.pcap -Y "ip.addr == 172.19.0.4 && tcp.port == 80" 2>/dev/null
```

7. Follow a conversation / stream

GUI

Right-click a packet > **Follow** > **TCP Stream** (or Ctrl+Alt+Shift+T). This reassembles and shows the full application-layer conversation, exactly what the attacker typed or downloaded. The best way to see a whole HTTP request/response as one readable block.

CLI

Export a conversation as a readable stream:

```
tshark -r /work/network.pcap -q -z follow,tcp,ascii,1 2>/dev/null
```

The 1 is the TCP stream index (start at 0). Use `follow,tcp,raw` for the hex/ASCII form.

8. Extract statistics for a report

Statistics make findings reportable and quantifiable.

Top talkers

```
tshark -r /work/network.pcap -q -z endpoints,ip 2>/dev/null
```

TCP conversations

```
tshark -r /work/network.pcap -q -z conv,tcp 2>/dev/null
```

HTTP requests (tabular view for reports)

The cleanest way to list every HTTP request with its source and URI, using field extraction:

```
tshark -r /work/network.pcap -Y "http.request" \
-T fields -e frame.number -e ip.src \
-e http.request.method -e http.request.uri 2>/dev/null
```

This prints one row per request - ideal for a report table:

```
226 172.19.0.4 GET /admin
238 172.19.0.4 GET /admin/index.php
251 172.19.0.4 GET /config.php
263 172.19.0.4 GET /backup
275 172.19.0.4 GET /backup.zip
287 172.19.0.4 GET /phpmyadmin/
299 172.19.0.4 GET /wp-login.php
311 172.19.0.4 GET /server-status
323 172.19.0.4 GET /robots.txt
335 172.19.0.4 GET /files/1.txt
347 172.19.0.4 GET /etc/passwd
359 172.19.0.4 POST /login
```

HTTP request statistics

Aggregate the requests per host and URI with the `http_req,tree` statistic:

```
tshark -r /work/network.pcap -q -z http_req,tree 2>/dev/null
```

This reveals request patterns, including the `/pixel.gif?id=beacon&c=N` C2 beaconing observed in the lab capture.

Save a filter-based export for evidence

```
tshark -r /work/network.pcap -Y "http.request" -w /work/http-requests.pcap 2>/dev/null
```

9. Search for indicators of compromise (IoC)

Use tshark to scan the whole capture for suspicious strings, such as a command injection payload, a username from a brute force, or a known-bad IP:

```
# search packet payloads for a string
tshark -r /work/network.pcap -Y "frame contains \"admin\"" 2>/dev/null

# search for a specific IP
tshark -r /work/network.pcap -Y "ip contains \"172.19\"" 2>/dev/null

# search raw HTTP for a credential or path
tshark -r /work/network.pcap -Y "http contains \"passwd\"" 2>/dev/null
```

In the GUI, use **Edit > Find Packet** (or Ctrl+F) with the filter bar to search the same way.

10. The GUI workflow at a glance

1. **Open** the capture (File > Open).
 2. **Colour** suspicious traffic: View > Coloring Rules (SSH brute force, HTTP errors, etc.).
 3. **Filter** to the attacker: ip.addr == 172.19.0.4.
 4. **Follow** the TCP stream to read the full request/response.
 5. **Add a comment** (right-click > Packet Comment) to annotate evidence.
 6. **Export** your evidence: File > Export Packet Dissections, or keep a filtered copy of the pcap.
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Quick reference

```
# list interfaces
tshark -D

# live capture
tshark -i eth0 -w /work/live.pcap

# read a file (one line per packet)
tshark -r /work/network.pcap

# count packets
tshark -r /work/network.pcap 2>/dev/null | wc -l

# protocol hierarchy
tshark -r /work/network.pcap -q -z io,phs 2>/dev/null

# filter by host and port
tshark -r /work/network.pcap -Y "ip.addr == 172.19.0.4 && tcp.port == 80" 2>/dev/null

# only HTTP requests
tshark -r /work/network.pcap -Y "http.request" 2>/dev/null

# endpoints (top talkers)
tshark -r /work/network.pcap -q -z endpoints,ip 2>/dev/null

# TCP conversations
tshark -r /work/network.pcap -q -z conv,tcp 2>/dev/null

# follow a TCP stream (index 1)
tshark -r /work/network.pcap -q -z follow,tcp,ascii,1 2>/dev/null

# list HTTP requests as a table
tshark -r /work/network.pcap -Y "http.request" -T fields -e frame.number -e ip.src -e http.request.method -e http.request.uri

# HTTP request statistics by host/URI
tshark -r /work/network.pcap -q -z http_req,tree 2>/dev/null

# search payloads for a string
tshark -r /work/network.pcap -Y "frame contains \"admin\"" 2>/dev/null

# export a filtered subset
tshark -r /work/network.pcap -Y "http.request" -w /work/http-requests.pcap 2>/dev/null
```

Pitfalls

- **GUI vs CLI binaries.** In the CPIA analyst container only the CLI (tshark) is installed by default. The GUI (wireshark) needs a desktop environment and a display. On a headless server use tshark.
- **Root warnings.** tshark prints "Running as user root ... dangerous" - this is informational. It still works.
- **Noise from 2>/dev/null.** Some tshark versions emit SSH dissector warnings for unusual key-exchange algorithms. Redirect stderr (2>/dev/null) to keep output clean, or use -q for statistics.
- **tshark -q vs -Y.** -q suppresses per-packet output (used with -z statistics). -Y applies a display filter to the packet list. They do different jobs; combine them as shown above.
- **Stream index in Follow.** follow,tcp,ascii,N starts at stream 0. Use the stream number you see in the GUI packet details if matching a specific conversation.

- **Time zones.** pcap timestamps are usually UTC. Note the zone when you quote event times in a report.